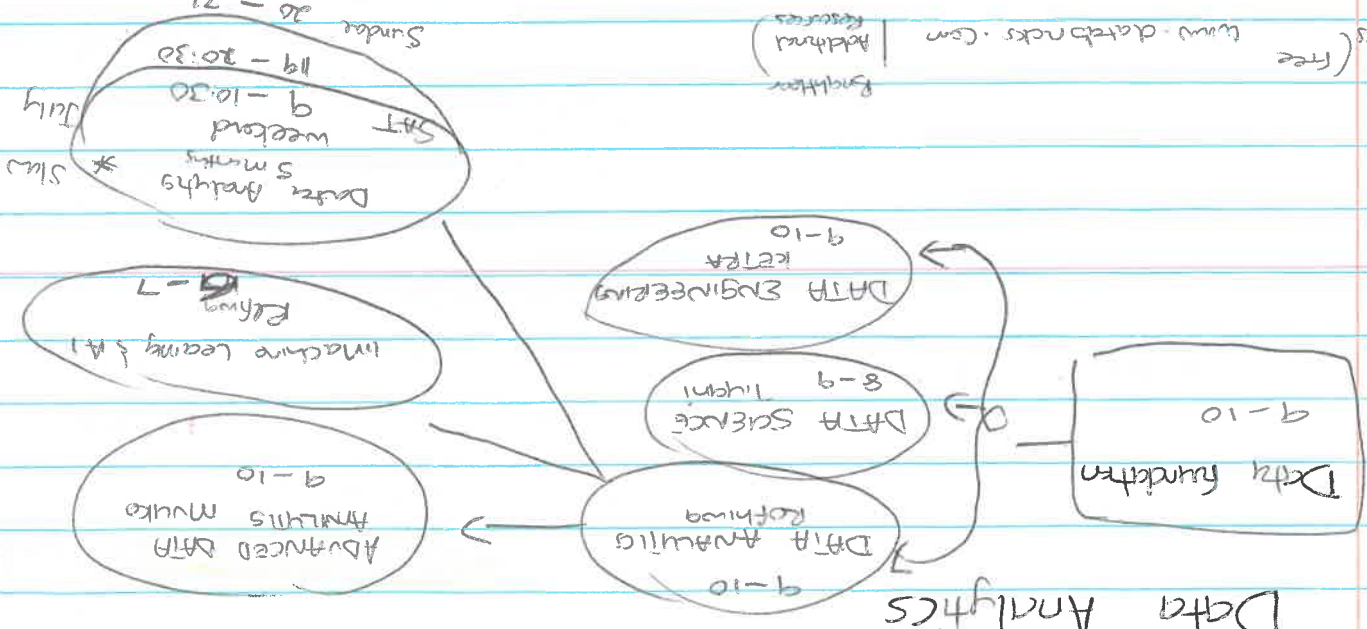


MASTER 10 DATA ANALYTICS

1/6/25

Data Analytics



DataSnacks (free) www.datasnacks.com | Additional Resources

2/06/25

Data Foundations

Info to Databases

- ③ Microsoft credentials
- ③ Azure fundamentals beginner
- ③ GitHub (Portfolio) Sign up
- ③ Google drive
- ③ Presentation Deck
- ③ Excel
- ③ First Tables
- ③ Analysis
- ③ Google Sheets
- ③ AI Tools - Smart AI, Power BI, HHS / Knowledge

- 7 Micro.com
- 8 Data Studio
- 8 Similar to Power BI
- 8 Cloud based Data visualization tool
- 9 Google Cloud Platform SCP
- 10 Coding
- 10 Data Repair
- 10 Data cleaning
- 10 Using SQL in DataBricks
- 10 Code -> Dashboard
- 10 Data Mgt System
- 11 Python code in data bricks
- 12 Google collab

13 draw.io (Architecture diagrams) <https://app.diagrams.net>

Data foundation

- is time factor (information is time based) -> to drive business decision
- is a collection of facts
- is information
- is information organised

Data is everywhere

Various Information - Voice

- Singing
- video

Types of Data

Structured data

- refers to data stored in a table with rows & columns
- has a consistent format

Row 1	col 1	col 2	col 3	col 4
Row 2				

eg Spreadsheet (excel file)

Name
 Surname
 Age
 Location

ID	Name	Surname	Age	Location
1	Ali	Mustafa	22	JHB
2	Issa	Medina	33	CPT
3	Yusuf	Mustafa	20	PSA

Backend stored like this

- eg music
- Podcast
 - Social media Post
 - Rich text
 - video
- eg music
- Podcast
 - PDF documents
 - Website
 - Youtube videos

- it has no specific format
- it is inconsistent format

unstructured data

Data - is a collection of facts

Data is everywhere

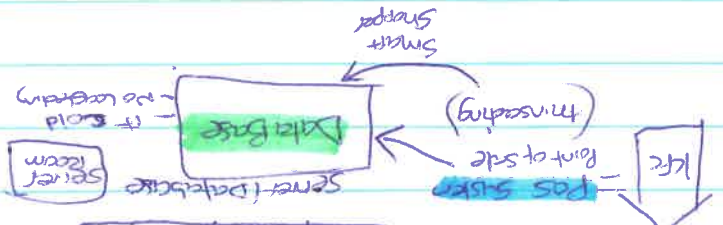
Banking Apps

eg Student ID

Data collection

Order ID	Product	Revenue	Date/Time
101	Straw	40	22/06/20
102	Ice Cream	5	
204	Apple	80	

Data collection -



- Data driven decision - is based on information based on the past
- To unlock decision
- To unlock future & potential of utilizing information
- To unlock customers
- To unlock business
- Patterns are monitored
- What relevant

Shop

Why certain shops close / operate; its previous



What drives this?

Location	Date Time	Revenue	Promotion Cell free	Buy 1
----------	-----------	---------	---------------------	-------

Job

Data Flow / Architecture / warehousing

Online Ingested Processing Database (OLTP)

Cloud

Snapshot

Message

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

data

Snapshot

allows Base
to data
used to capture
data but
for processing
- it is not used
complex
queries
- does not
allow

allows Base
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for processing
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complex
queries
- does not
allow

Data Engineering

Data Science

Data Analyst

Business Intelligence

Machine Learning

Report

Collect, more / store

organising data

Dashboard

is available, secure, reliable

hard system that makes data

Analysis

Source System

ingest - (Data Storage)

Source System

Source System

Source System

Source System

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Source System

Insurance

Data Warehouse

Report Analysis of Message

Share

Buy

ETL

Warehouse

Report

Analysis of Message

Share

Buy

ETL

Warehouse

Report

Analysis of Message

Share

Buy

ETL

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ETL

Warehouse

Report

Analysis of Message

Share

Buy

ETL

Warehouse

Report

Analysis of Message

Share

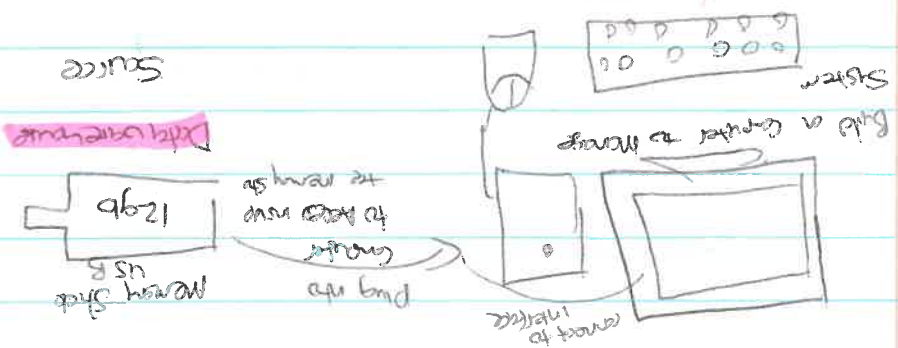
Buy

Mat of "DMS"
Data Bndt
w/erter
manage dms

Data Engineering

Data Ingestion - from its source into main system
Zentraler github
create
Storage
Azure

Data mgt System



Data Mgt System - is

an interface
is a system that gives access to our data

oslopps

Data Visualisation

(Start Talking)

1. How Revenue over the years



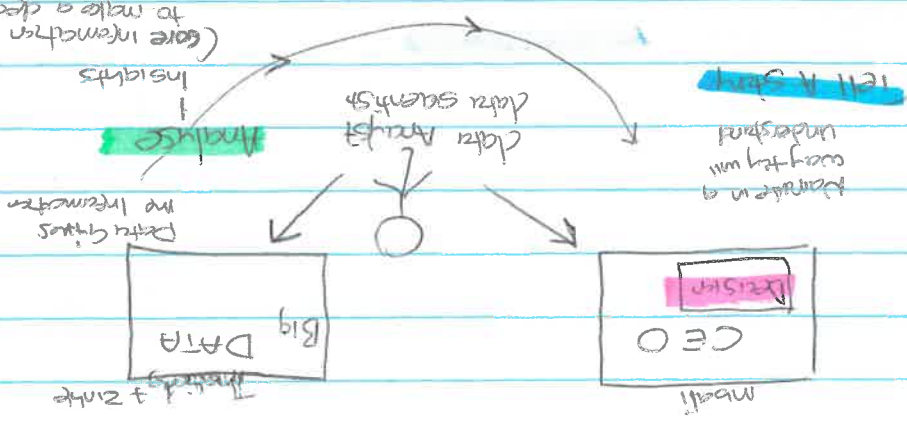
- Data is information of (Thinking + Zinke) holding friend
 - (Tell me!!) what is going on (make this story)
 - (mobile) will take a decision - breaking

- Take the data & narrate the story to the company

- You need to be able to tell a story from your data

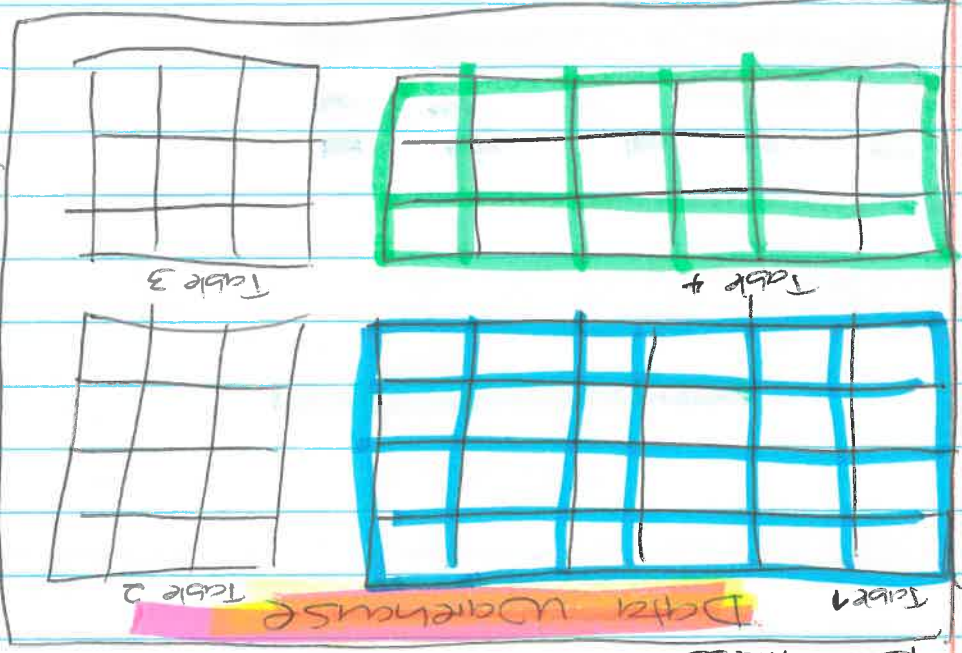
- You need to drive / extract actionable insights from your data

- This is to drive business decisions



Data driven
 Recommendation
 Data insights
 Diagnostic becomes

Data Process



- Data Warehouse is
 - is something that stores data
 - data that can be related/joined
 - together to become 1 eg
 - Dimensional tables / fact tables
 - Reasons why it's called
 - Relational data base
 In a Relational database we
 store structured data in
 dimension table & fact tables
 fact can be compared to 1
 dimension using
 primary key & foreign key

Dimension Table

ID	Name	Age	City
1	Rishi	28	NY
2	Jeff	22	LA
3	Mike	18	Michigan

Fact Table

ID	Category	Product	Value
101	Fruit	Apple	102
102	Fruit	Apple	103
103	Electronics	Laptop	104
104	Electronics	Laptop	105

Dimension Table (Dim)
 Fact Table

- is just a table that just give
 a description table
 - information that does not consistently
 change continuously
 - Dim Table contains descriptive information
 ID is
 Primary key & Customer ID
 is a unique identifier that identify/differentiate
 each component from the table
 * Concept of foreign keys is ID

eg Home Affairs

Primary key - help us to identify each row differently eg. ID No., @ Home Affairs

- Barcode
- Unique No.

- contains different ID from different tables
- it only contains words no words
- A fact table has foreign keys
- contains rows used for analysis

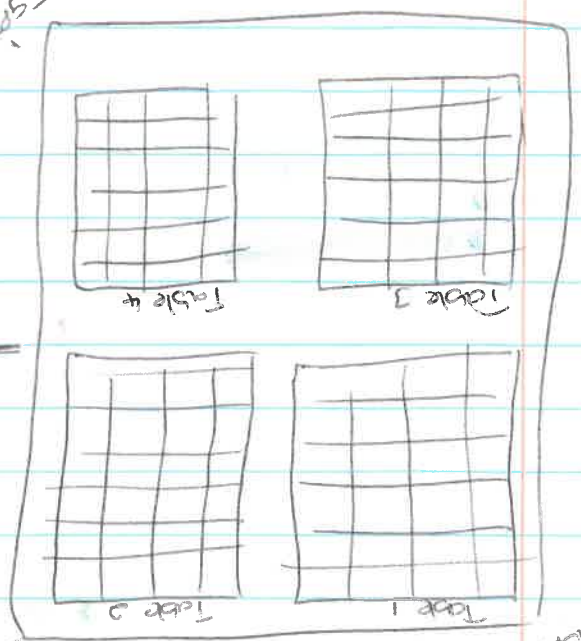
Transactions

Order ID	Customer ID	Products	Revenue
234	2	104	70
333	1	102	80
401	3	101	100

- Foreign key** - is a primary key from 1 table that is found in another table
- A place where you don't belong
- A customer ID exist in another table

Data Process

Data Warehouse (Storage meaning system) and



Why do we need to code?

- Data Inspection** - to see what data is available
- Data Exploratory Analysis** - to understand the relationship between our data points
- Find Relationships** (Business rules is related to another table)
- Is better, Manipulate, Amend, update, clean, integrate** to join, to delete, update

Preparing/structuring database / Archive insights

- future building for machine learning for predictions
- creating features used in ML for predictions

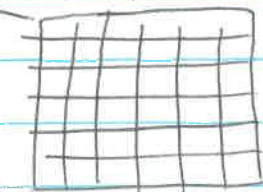
"DATA CHIEF"

End up with a Summarised Table on aggregated table / combined / joined table

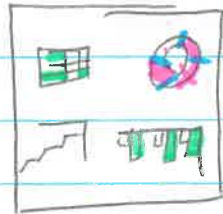
Want information that I want to turn

Unified Table

"Insight Table"



Power BI Reports Reporting Tool



Machine Learning

Prescriptive Tool

Unified Table

Code

Data Mgmt

Data Warehouse

DATA PROCESS

clean! Summarise type of table to use how to tell story

Why need visuals

How did it come? need to use the draft

Why the chart? (Lump graph scatter plots)

Scatter graph line / Area chart

STEP 1

BRS - Business Required CEO
- I want to understand different products for the shop for different products
"The story"
"The why"
Reason Behind

Business Requirement
Data driven question to understand
Processing the data - what do
Visualise
Insights



Age brackets
Should I focus on specific categories?
Do you want to compare
Summarised?
Do you want to use 13 years with VAT inc / Excl?
Analyse frequency of purchases
Repeat customers? to
Would you like to see a
like to see daily revenue
Which products would you
Ask data driven?
Do you want to ask
Data driven questions
"Data Analyst"
"Business understanding"
Check-SPJ

3 CODE (Data Processing)
4 Data Analysis
5 Visualisation
6 DATA PRESENTATION
Putting things together
Insights generation

clean → Data Grids
Data Studio

Stack chart

Bar graph → Show proportion and different categories by
→ comparing
- Product category - Beauty, deodorant, cloth
- Day of week - Day of week, Mon, Tue, wed



Excel & Data Studio

Practical
Punch card sale
Daily income in coffee shop
Jan

Processed data
Purchase date
Date of sale
Unit of sale
Unit of product
Product detail
Store location
Click top left, click with product category
Click Data
Click on label
First tab
Click ok

Filter
Months

Insights
We are asked to interpret
to data instead of
another values

① Purchase date
Revenue trend graph
② Revenue day
③ Store Name
Click ok

Delta Studio

Create
Click into to data
Build like column

Revenue
Sales
Products

Change connectors - CSV file upload
- create a dataset
- continue
- coffee background

Scroll down

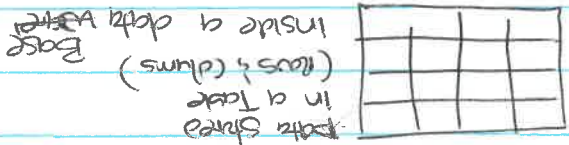
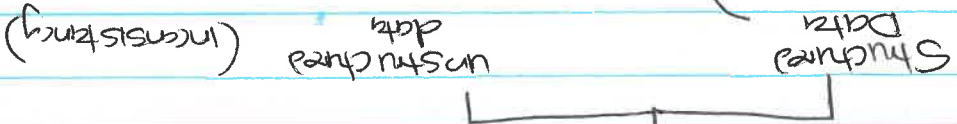
Add data
Chart - double tap
Add new or a dashboard
Add a chart
Setup style
- Choose your unit of sale
Metric - unit of sale
- dimension - category detail
- location
Style - Include chart title - Revenue Dashboard
- by store location
Uniform should be equal

SQL

Introduction

- Practice Everyday 30min consistently
- Make it a routine frequently
- Accountability (Learn from People)

Data Information we use to make business decision



Inside a data table Base

Make business decision

When we have data we need to Extract Insights from the data

We need a programming tool to extract the data

Structured Query Language (SQL) → is a programming language we use to

Interact/extract / retrieve information from a structured table

→ We use SQL to Retrieve data/Information from a structured table



Syntax

STUDENT Table Results

Std ID	Name	Marks
101	Rafhi	99
102	Mbali	85
103	Thabo	50
104	Lehtabo	60

SELECT name, marks

FROM

Student Result

Table

NAME	Marks
Rafhi	99
Mbali	85
Thabo	50
Lehtabo	60

SELECT Statement

- Allows to select specific columns from the table

FROM Statement

- a statement allows to specify the name of the table you are selecting from

SELECT * - wildcard
Set entire

SELECT * - is a way to select all columns in a table without specifying the column name by 1

SELECT * - is to get all the column names in that table

Retrieving

2

SELECT

date,
gender,
Age,
only

FROM

retail - sales - dataset; (Name of the table)

choose pipe
* Separate each column name by a comma
when selecting specific column names
- last selection you don't put the (,) as after only

3

SELECT DISTINCT STATEMENT (Different)

You want to get different values

from a specific column

- unique / different
- We remove duplicates / we don't repeat
the same value

Syntax

SELECT DISTINCT

column - name

FROM

table - name;

Product ID	Product	Order ID
101	Apple	100
102	Orange	50
103	Apple	60
104	Banana	150

Fruits - orders

order - id	Product name	Revenue
101	Apple	100
102	Orange	50
103	Apple	60
104	Banana	150

SELECT DISTINCT product - name

(Specify to column name
you want different value
from)

FROM Fruit order



Product Name
Apple
Orange
Banana

output

SELECT DISTINCT Product category
FROM retail - sales - dataset;

3 ways to select

1. simple

SELECT *
FROM table;

- is a query
selecting all the
columns from our
table

2.

SELECT DISTINCT

- Specify to column
Name want different
value in specific column
- no duplicate

3.

SELECT

column - name

FROM table - name;

Want to select
Specify to column
we want

"ALYX THE ANALYST" Data Bkts

- Dint water form woodlies - it's filters
- it doesn't contain backing

FILTERING STATEMENT

Data filtering - It is a way for us to filter or get specific information from data

- we filter data if we want to get specific information from our data
- from 2006 Jan - Dec
- "I don't want everything"
- Ex: Just data based on specific filters
- Is a way to exclude rows of data that don't need a specific criteria
- You want specific information
- We filter data based on conditions

WHERE STATEMENT Syntax

SELECT
column 1,
column 2,
column 3
FROM
tablename
WHERE
Column condition

filtering

Student Name	Marks
101	Rishi
102	Mahi
103	Thad
104	Ad

Student - Result

Write an SQL query to give me only the name of student that obtained more than 60% in their exam

Where marks > 60%

Answer
101 Rishi
99

Filtered table
Reduced table

Multiple column (Comparison operators)

SQL operators

"CONDITIONS"

10/06/20

Arithmetic operators

Operator	Description
+	Addition
-	Subtraction
*	Multiplication
/	Division

Comparison operators

Operator	Description
=	Equal to
>	Greater than
<	Less than
>=	Greater than or = to
<=	Less than or = to
<>	Not equal to

Logical Operators

Operator	Description
AND	both conditions must be met
OR	one condition must be met
BETWEEN	Must be within the range
IN	Must be contained within in the list
NOT	Gives the opposite result of what is specified
LIKE	looks for a pattern

Operator is a tool that is used to do a lot of work (multipurpose tool) to build to conditions

Data Filtering - a we use the WHERE STATEMENT

It is a filtering statement that we use to filter our data based on specific

Conditions

- We can use the SQL operators to build the conditions for filtering our data

Syntax - is the way you structure the code

```

SELECT
column 1,
column 2,
column 3
FROM
table name
WHERE
(condition)

```

Example

User ID	username	city
1	Namunliq	JHB
2	Daniel	CPT
3	Amele	DBN
4	Keselo	PTA
5	Lerato	PE

column name

Users

values within column

```

SELECT *
FROM users
WHERE city = 'Durban'

```

↑
Amele DBN

User ID	username	city
3	Amele	DURBAN

Link SQL Query to give me the name of users in DBN

used operator to

build to condition

15/06/20

Logical operators

AND STATEMENT

- is a statement that allows us to combine multiple conditions
- All the conditions must be satisfied
- We commonly use the AND statement when building logical conditions for filtering our data

SELECT

SELECT (column 1, column 2, ...)
FROM table - name
WHERE (condition 1 AND condition 2 AND condition 3...)

- no limit
- specific conditions
- multiple conditions
- in conditions

STUDENTS

Student ID	name	Course	Age	city
1	Thabo	Data Analysis	22	JHB
2	Sarun	Data Science	19	PTA
3	John	Data Analysis	25	D&N
4	Leato	Data Analysis	20	JHB
5	Musa	Data Science	23	JHB

1. Write a SQL query to give me the name of students living in JHB that are doing data analytics

SELECT *
FROM Students
WHERE City = 'JHB' AND Course = 'Data Analytics'

Membership plan

plan ID	plan-name	monthly fee
101	Basic plan	79
102	Standard	129
103	Premium	199
104	Family plan	249
105	Mobile plan	59

Monthly fee >= 150
150, 151, 152

1. Give me the monthly

Write an SQL query to give me the plans that cost more than 150

SELECT *
FROM membership-plan
WHERE monthly fee > 150

plan name	plan ID
premium	199
family	249

OR Statement - combines multiple conditions

- 1 of the conditions must be met
- at least must have 1 (car or Apartment)
- (car or share car)

Write a SQL query to give me the name of students living in JHB or student doing data analytics

Select *
from Students
where City = JHB OR Course = Data Analytics

Answer
LEOATO
THABO
MUSA
JOHN

"At least met 1 conditions"

IN Statement - It is a way of filtering data by specifying the values of interest from a specific column

- Specify infinite values in the column
- No limit in the number

Syntax

Select *
from table-name
where column in (value 1, value 2, value 3)

Write an SQL query to give me people living in Durban and Pretoria

Select name, city
from - students
where city in ("Durban", "Pretoria")

Answer
John & Sarah

NOT IN

- is an exclusion statement
- Write a SQL query to give me people not in Durban
- Returns
- it gives the opposite result

Select name, city
from students
where City NOT IN ("Durban", "Pretoria")

Answer
THABO, LEOATO, MUSA

BETWEEN Statement

- It allows us to filter the data based on a specific range of values
- It is an inclusive statement this means that the lower and upper bounds of the between statement are included

Select (column1, column2, column3)
from students (table-name)
where column name BETWEEN lower bound AND upper bound

- 1. Write an SQL query to give me the name and age of students between the ages of 20 to 23

Select name, age
from students

where Age BETWEEN 20 and 23

THABO, LEOATO, MUSA
20 23

"I have boundaries"

" I will sort you out "

Sorting

- is a way to sort our data in different ways
- there are 2 ways to sort ASCENDING or DESCENDING ORDER
- Sorting does not remove or add
- Order BY statement
- sorting new into our data, it just rearrange the rows in our data

- In SQL this is a sorting statement

Ascending order - it is going up from smallest to biggest
 (ASC)
 ↓

Descending order - it is going down highest to smallest
 (DESC)
 ↓

SELECT *

FROM table-name

WHERE condition

ORDER BY column DESC;

SELECT *

FROM table-name

WHERE condition

ORDER BY columnname ASC;

Student

Student ID	Name	Course	Age	City
1	Thabo	Data Analy	22	JHB
2	Sarah	Data Science	19	PTA
3	John	Data Analytic	25	DBN
4	Lerato	Data Analytic	20	JHB
5	Musa	Data Science	23	JHB

SELECT name, Age

FROM students

ORDER BY Age ASC;

ORDER BY age DESC;

Answer

Name	Age
Sarah	19
Lerato	20
Thabo	22
Musa	23
John	25

Name	Age
John	25
Musa	23
Thabo	22
Lerato	20
Sarah	19

Single " you are aggregated by group

Aggregar functions

Is used to perform ~~on rows~~ a calculation on rows of a specific column and returns a single value / a single value per group

function	Description
1 Count () (count (*)	Counts the number of rows count to all the rows in your table
2 Sum ()	Calculates the total of a numeric column
3 Avg ()	Calculates the average of a numeric column
4 Max ()	Finds the maximum value
5 Min ()	Finds the minimum value

Employees

Emp ID	Name	Salary
101	John	20
102	Mary	40
103	David	35

Select min (salary) AS min-salary
from employees;

Min

Min - Salary
20

Answer

Select Max (salary) AS max-salary

Max

Max - salary
40

Answer

Min
SELECT MIN (column-name)
FROM table-name
WHERE condition

Syntax

MAX

SELECT MAX (column-name)
FROM table-name
WHERE condition

AS Statement - we use it to rename a fresh name

Alice → Anisi
Alli

AS is called an Alias
is a nickname
- is a way to give rename a column name a descriptive name after applying a function.

Category	Total Revenue
Electronics	270
Food	80
Clothing	150

FROM Sales
GROUP BY category

SELECT category,
SUM (Revenue) AS total-revenue

Ans

Sales ID	Customer Name	Category	Revenue
1	John	Electronics	120
2	Mary	Food	80
3	Elon	Electronics	150
4	Peter	Clothing	100

Sales

- We group by a column that is not aggregated in the select statement
- we commonly group by a categorical column

GROUP BY subject

FROM STUDENT

SELECT COUNT (DISTINCT Student ID) AS
Roll of Students (Aggregated column)

Subject - not aggregated categorical column must be in group by

- Is a statement that allows us to group our data into categories by subject (the many are doing science, commerce, history)

GROUP BY Statement

- we count something that is uniquely identified like an ID
- we do not sum an ID rather we count on ID
- we do not count revenue we sum the revenue

Roll of Student
6

FROM STUDENT

Roll of Students

SELECT COUNT (DISTINCT Student ID) AS

- In SQL we can count a group

Student ID	Name	Subject
101	Rafael	Science
102	Jeff	Commerce
103	Steve	History
104	Marc	Commerce
105	David	Science
106	Elon	Science

(6)

STUDENTS

Student ID	Name	Subject
101	Rafael	Science
102	Jeff	Commerce
103	Steve	History
104	Marc	Commerce
105	David	Science
106	Elon	Science

- counting different student in the class by using their unique identifier

SELECT SUM Revenue
FROM Sales
GROUP BY department

Q Write an SQL query to calculate the total revenue for each category

INTRODUCTION TO DATA BRICKS

coding
SQL
Python
dashboards

New Tab from your google

first link

Try Databricks

Sign up (google account / continue with google)

For Personal use - get free Addition

- location
- continue
- puzzle
- submit

Code 30 min/day * very important

Workspace - link databricks → Github (Referral)
link in the

- Recent's - your work

Catalogue - very important

SQL Editor - opens all worksheets where u are editing stuff

Queues - think of same as Recent (SQL Split only)

ABC - test
123 - big ins

CATALOGUE

Go to Bucket

Under Additional Resources

- dataset - download

upload dataset

Home

Catalogue

Click +

Create a catalogue

Catalogue name - Retail - use default storage

Create →

- Select Table
- Browse
- select dataset downloaded from Retail / Reserve it

Review of the table

Retail
Database
Schema
Tables

* - we don't want AI suggested description

Create Table @ the bottom

if can upload multiple table in a schema

Sales



click schema
create

click dropdown

table

button

remove or table

create

if information is not / messy not a table

what do I do? eg CSV file

* Advanced Analytics

Column delimiter etc. tabs

No save / anything just close

Catbridge

Debridge

* Rename
New click
double click

Boast: shd ticks under ESC

Coding

Select for dataset (table

- Top next (create)

- Dropdown

- query

- first query is given for free
sorter & know what set

- Click: Run all or Ctrl enter

Aggregate functions

- Summarise data to extract specific information
- Perform calculation on multiple rows of a specific column & returns a single value / single value per group
- It consolidate

Count () - we count unique identifiers (id)
 Sum () - we sum numeric data (Revenue)
 Avg () - we average numeric data (Revenue)
 Min () - looks for min value in a column
 Max () - looks for max value in a column

GROUP BY Statement

Select (categorical column, (gender, race, category) with words)
 Aggregate - function (column) AS describe column = name
 FROM table - name
 Group by (categorical column);

We group by a column that is not aggregated

HAVING Statement

- It is used to filter data that is aggregated
- We apply the HAVING Statement to data that is already aggregated
- It is used to filter grouped results after using a GROUP BY

WHERE Statement filters rows before grouping
 HAVING Statement filters groups after grouping

If a column has Avg (Pne) - its aggregated

But if we add a Product - name
 then it groups by product - name

Custom - id	Customer name	Product	Amount
101	Rafhi	Laptop	5000
101	Rafhi	Mouse	300
102	Abel	Keyboard	700
102	Abel	Monitor	2500
103	Clara	Mouse	300

Sales

Q Write an SQL query to find customers who spent more than 3000

Select Customer name, Aggregate Sum (Amount) AS total-spent From sales Group By (Customer name) Having Sum (Amount) > 3000

Customer Name	Total spent
Rafhi	5300
Abel	3200

SCENARIO

SELECT	Customers
Aggregate	Sum (Amount) (Amount with nulls)
AS	Total spent
From	Sales
Group	Customer
HAVING	Sum Amt > 3000

LIMIT Statement

Alter anything - does not change anything it just reduces the no. of rows in final output

It is a way of restricting the number of rows you want in your final output

ID	Name
101	Rafhi
102	Misali
103	Zuma

Employees

SELECT * From employees Limit 2

ID	Name
101	Rafhi
102	Misali

Ans

Kaggle
 # Project
 # Bright TV sales
 # Bright eye shop

branch of datasets - Kaggle

GITHUB

- is a platform to be used to store all the work (Portfolio)

- is for version control

- Store, Manage, Edit & collaborate on software code

Go to google - type Github.com
 Be Public about your journey

- Sign up
- connect with google
- Create account

Open user navigation Profile

Click on the profile

1

Name - Put your name & surname

Bio

Data Analyst currently sharpening my skills in SQL, Python and visualizations using Excel, PowerBI, R, Python, data bricks and Data Studio

Social Media

1 LinkedIn

Location JHB

update profile

Click on the Name to see if its updated

Pushing something to work, "commit & Push" - it puts a dot on the calendar like

Click on the link of Github & email to whoever Portfolio goes to

Submit your work

First thing

Blocks are called Repository on the right
Repository is a folder

Section 1

How to create Repository

+ Sign or create Repository

Click on Top left Repository

Click new

Name the Repository Name : SOL Fundamentals Exercises

Description : This repository hosts some exercises done on basis on SOL

Section 2

Language - Public (always its on Public)

Submit for exercise

Add README - ON (always on)

Click on your Github link
Send email to Github!

Add Githubignore - NO

Add license - NO

Create Repository

How to upload (Handwritten) Scan it & save on computer

Click Add file

Upload file

Choose your file

(Commit Changes (means save)

Short comment (uploading file)

Click commit changes

upload different file

Add file

upload file

PDF

file suppose to be new

Commit
Change

uploading file into folder
(commit change)

(commit change)

Choose a file

Upload file

Add file

Upload

(commit message creating folder)

(commit change)

eg TEST FOLDER / placeholder

Give folder name (TEST folder) / if its blue its a folder

Create new file

Add file

New folders

Case Study Shrike

(commit changes)

A1 message (already written for u)

(commit changes)

delete file

3 dots...

Click on file

Delete (made in) over after ~~later~~

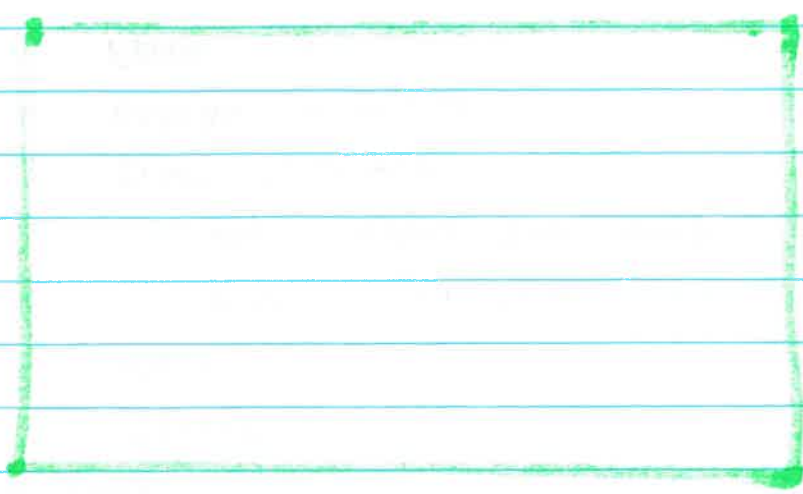
Creative

Reactive

- A description of what Project Overview Case Study directs Dataset Notes

You can use AI to help

Delete Repository
Settings
Danger Zone
Delete Repository
Via Email
Send code



Setting
Send down to danger zone



Go in Repository

CASE STATEMENT

- Is a way to build logical conditions
- If allows to create a new column based on logical conditions applied on other columns
- Is a way to allows us to classify our numerical data into categorical groups
- Take something that is numerical to make categorical column

Syntax

```

SELECT column1,
CASE
  WHEN condition1 THEN result1
  WHEN condition2 THEN result2
  ELSE default result
END AS column_name
FROM table_name
    
```

eg

Emp-ID	name	Salary
101	John	3000
102	Peter	7000
103	Samson	12000
104	David	4000
105	Gilgith	800

Employees Numerical

Classify term low, medium, high Salary

Salary below 8000 = low
Salary btwn 8000 and 10000 = medium
Salary above 10000 = high

Logical conditions =

Business Logic

SQL logic

```

SELECT name, salary,
CASE
  WHEN salary < 8000 THEN 'low'
  WHEN salary BETWEEN 8000 AND 10000 THEN 'medium'
  WHEN salary > 10000 THEN 'high'
END AS salary-bracket
FROM employees;
    
```

Case statement is placed within the select statement

Case statements creates a new column

name	Salary	Salary bracket
John	3000	low
Peter	7000	medium
Samson	12000	high
Dani	4000	low
Goliath	300	low

Output

Exercice 2 Products

Product-id	Product-name
1	Laptop
2	Phone
3	Keyboard
4	Monitor
5	Mouse

Classifying each Product by Price

• "Expensive" if Price > 1000

Mid-range if Price between 100 and 1000

Budget if Price < 100

Sql Syntax

SELECT product-name,

Price

CASE WHEN Price > 1000 THEN "Expensive"

WHEN Price BETWEEN 100 AND 1000 THEN "Mid-range"

WHEN Price < 100 THEN "Budget"

END AS Price category

FROM Product-name

Product-name	Price	Price category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

Ans

length
matsys.

GITHUB TO DATABASES LINKING

①

Click m to next over

Setting

User - linked etc

Add git credentials

link git etc

Click link

Authenticate

link success

Click configure in Github

All repositories selected

password - verify code via email

Create

Repository on Github

Click on picture top right

Profile

Repositories

3 Copy into database
column = repository to 4

Click code

Click HTTPS - 2 Copy

Go to Databases

Click dark bricks to - Github
if it is too big it go -> Github

Click on 3 dots

(Pushing)

Click git

uploading test file (commit message)

commit & push

Github & Databases must be exactly in

github

Databases

coding file format

Click m

(locking)

Settings

developer

Default file for databases CH1 f

Source (py, sql)

Databases

coding folder

Click data set

(Folder)

Click create

notebook (Python)

Developer click - Retail - sales data coding

Click SQL

Confin

Click on square

Path

Select from Retail, sales, dataset

highlight it

click drop down

run selected test

Select Dishes gender from sales data dataset

Select it drop down
run selected test

left

Push to Github

Workspace

⋮ - GH

Commit message - use Backtick
Commit & Push
Successfully Commit

Switch off A1

to Github

2 words in your column name it must have backticks

* Product category should be underscore backticks

* The functions are not case sensitive
columnname - are not case sensitive

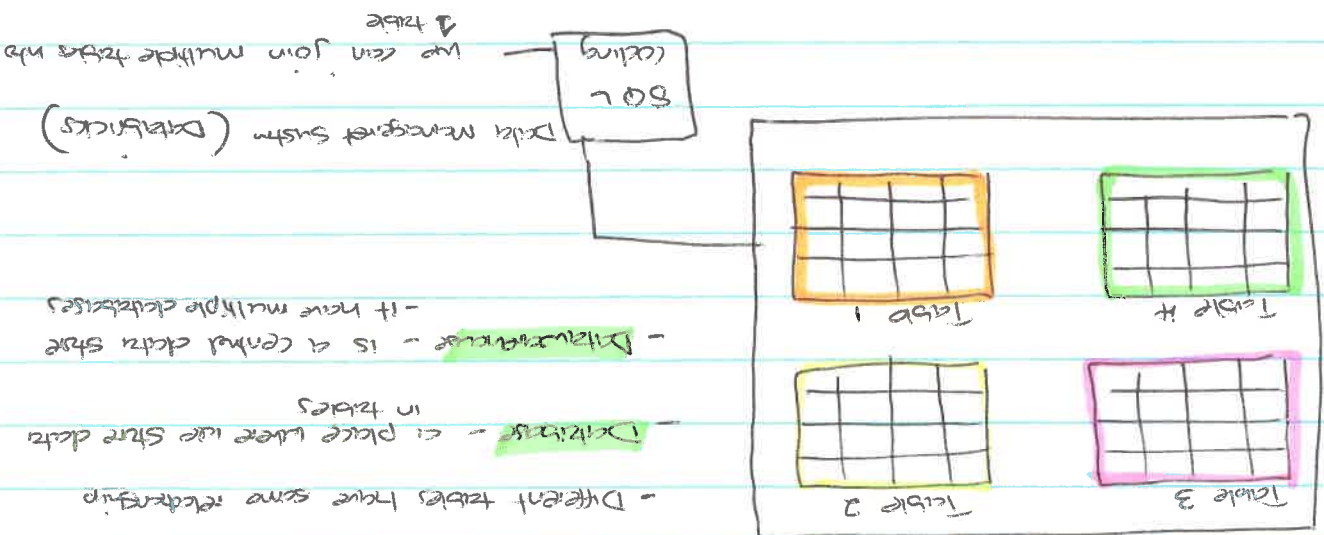
* Items inside the Table is case sensitive
"Brady"

SQL JOINS

Related database

In database we have multiple tables related to each other (related database)

- Different tables have some relationship
- **Distributed** - is a place where we store data in tables
- **Distributed** - is a central data store - it have multiple databases

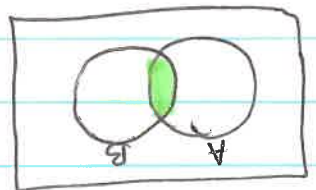


SQL JOINS

Share some problem of vector

- An SQL join is meant to join multiple tables based on common columns
- Common column is a common value found in both tables

INNER JOIN



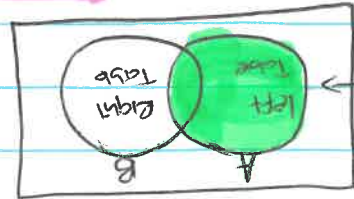
- We join tables on columns but the join affects the output rows
- When we join tables we can select any column between the tables we are selecting/joining (which means is mine)
- Inner join - Returns rows where there is a match on both the tables

Syntax:

```
SELECT
  column 1,
  column 2,
  column 3,
  column 4
FROM Table 1 AS A
INNER JOIN Table 2 AS B
ON A. (common-column) = B. (common-column)
```

2

LEFT JOIN



- Main table
- also known as
- Base table
- table of focus

Syntax

SELECT
column 1
column 2
column 3
FROM
table 1 AS A
LEFT JOIN table 2 AS B
ON A.common column = B.common column

- We have a left table & a right table
- We need everything in the left table in our final outcome
- It is our priority table
- I want everything in the left table, but from the right table I only take the things that matches

Order ID	Customer ID	Amount	Product ID
101	1	500	131
102	4	20	132
103	5	10	131
104	1	300	133

Product No.	Product ID
Apple	131
Mango	132
Orange	133

- Fact Table**
 - It's a fact table
 - no actual name of the product
- Dimension Table**
 - Descriptive table (it won't change everyday)
 - gives more information to the fact table
 - static descriptive information
 - data stored in a dim table does not change frequently
- Primary Key**
 - Customer ID is a foreign key in orders table
 - is a primary key in 1 table but foreign key in another table
 - it is not unique, it can be repeated
- Primary Key**
 - Customer ID is a primary key in the customers table
 - is a unique identifiers
 - customer ID is unique identifier for each customer

Customer ID	Name	Amount
1	Rohit	500
2	Mahi	10

SELECT A.Cust-ID,
A.Name,
B.Amount
FROM customers AS A
INNER JOIN orders AS B
ON A.Cust-ID = B.Cust-ID

Order ID	Customer ID	Amount
101	1	500
102	4	20
103	5	10

Customer ID	Name
1	Rohit
2	Mahi

• 1FC - 1st order shakespeare
2nd order 1400000
customer ID can purchase several times only
the order ID will change.

LEFT JOIN returns all records from the left table, and only the matching

records from the right table

When there is no match in the right table, we still keep the left table row, but the right table column will show **NULL**

left
BASE TABLE

Customers Table

3	mbali
2	leato
1	Rofhiwa
(cust-id)	(cust-name)

right orders Table

102	2	300
101	1	500
(order-id)	(cust-id)	(Amount)

Think an SQL query to give me a list of all customers and what they ordered

SELECT cust-id, cust-name, Amount, order-id
FROM customer-table AS A
LEFT JOIN order-table AS B
ON A.cust-id = B.cust-id;

Ans

3	mbali	NULL	NULL
2	leato	300	102
1	Rofhiwa	500	101
(cust-id)	(cust-name)	(Amount)	(order-id)

What is a NULL
 - is an empty space, a missing value

M1120

✓OK

Insight - is information that comes from Data
 - Actionable
 - Based on information I can do something

① Understanding the Business Requirement (Reading & understanding)

② Go to mind

Copy & Paste case - in Document

① Step 0 (Reading & understanding the Business Requirement)

Step 1 Asking data driven Questions

- Research & design

Step 2 - do we have the data to answer these questions

Inspect to data on different

Check to overall data

Step 3 Inspect data on different

Step 4 Check to available data

Step 5 data processing

after inspecting the data check what can be done from the data. This includes data cleaning, data manipulation

① Select Distinct gender

From workspace: default.user - m

gender check

② Select Distinct

CASE

when gender = 'none' THEN 'unknown'
 when gender = ' ' THEN 'unknown'
 when gender = 'is null' THEN 'unknown'
 ELSE
 when gender is gender sex
 FROM table -

SELECT DISTINCT RACE
 FROM user - profile
 CASE
 when race = "other" THEN unknown
 when race = "not" THEN unknown
 when race = " " THEN unknown
 ELSE
 when race is null THEN unknown
 END AS ethnicity/race
 FROM profile table
 GROUP BY ethnicity

gender data

Province checks

RACE checks

SELECT DISTINCT

CASE Statement

When gender = 'None' THEN 'unknown'

When gender = " " THEN 'unknown'

When gender IS NULL THEN 'unknown'

ELSE gender (Give me the gender as it is) If gender is M or F return as it is

END AS sex (new column name)

FROM workspace.default.user - profiles;

SELECT DISTINCT race

FROM workspace.default.user profile

SELECT COUNT (DISTINCT userid) AS sub

(CASE

When race = 'Other' THEN 'unknown'

replace with other

replace None with unknown

replace empty with unknown

replace Null with unknown

ELSE race

END AS ethnicity

FROM workspace.default.user - profiles

GROUP BY ethnicity

1 SELECT DISTINCT province

CASE when province = 'None' THEN 'unknown'

when province = " " THEN 'unknown'

ELSE Province

END AS Region

FROM workspace.default.user - profile

Age
Order

SELECT min(age) AS min-age,
max(age) AS max-age,
avg(age) AS mean-age
FROM workspace.default.user-profile
check the youngest person
check the oldest person
check around person

SELECT

CASE WHEN Age = 0 THEN 'infant'

WHEN Age BETWEEN 1 AND 12 THEN 'kids'

WHEN Age BETWEEN 13 AND 17 THEN 'youth'

WHEN Age BETWEEN 18 AND 35 THEN 'young Adult'

WHEN Age BETWEEN 36 AND 50 THEN 'Adults'

WHEN Age > 50 AND Age <= 60 THEN 'Elderly'

WHEN Age > 60 THEN 'Pensioner'

END AS age-group

FROM workspace.default.user-profile;

* Combine All under 1 select Statement

Check for Duplicates

Select Count(*) as cnt,

user id

from profile-user-profile

group by user id

having Count(*) > 1;

- ViewShipday
- agent Chat - by Monday
- mind

Project
Guan

6/7/26

DATA PROCESSING
DATA VISUALISATION
DATA PRESENTATION

MIND

CANVA

7-8 pm
7-8 pm
7-8 pm

7/7/26 8-9 - consultation with Leanne
tools to be used
Sunday 8-9 pm
Data report
Project engineering

operational efficiency project
gantt chart

Canva - Project plan

Search - Project plan gantt chart
+ status update
Bright

Bright
Bright core, snap sales model

(PDF FILES)

Bright TV Version Analysis

live live 01 July 07 July 14 July 28 July 04 Aug 17 Aug 25 Aug

Business Requirements

DATA DISCOVERY

Defining problem space
collaborative

Business understanding - comparison
- media with data
Defining success measures

②

DATA DISCOVERY

Data inspection
Data exploration Data analysis

③

DATA PROCESSING

DATA PROCESSING
Data cleaning
Feature engineering
Using algorithms
Summarising

④

DATA ANALYSIS VISUALISATION & PRESENTATION

DATA ANALYSIS VISUALISATION & PRESENTATION
Insights extraction
Summarise insight / creating a dashboard
Compiling presentation

⑤

DATA PRESENTATION

DATA PRESENTATION
Monthly view of daily revenue
Update new products into report

Data Basics

Data Types

①

SELECT

I want to see what is in a table

FROM - workspace . default . viewshop

LIMIT 10;

} .

USER_ID → is an integer (INT)

(NUMBER WITHOUT A DECIMAL)

Channel color → contains String

records + numbers together combination of multiple characters

STR/object - is a combination of characters A-Z a-z 0-9 * ? ,

DATA TYPES

which are data types?

INT (integer) is a number without a decimal (078725)

STR (string/object) is a combination of characters (a-z / > ? ,)

DateTime, Timestamp

DATE FUNCTIONS

- help us to deal with date columns . manipulate to date columns .

SELECT TO_DATE (record2) AS water_date - converts a string into a date yyyy-mm-dd

FROM workspace . default . viewshop;

SELECT

Record Date 2,

TO_DATE (Record2) AS water_date,

converts a string into a date

extract to day from

DAYNAME (TO_DATE (Record2)) AS DAY_NAME

extract to month name

MONTHNAME (TO_DATE (Record2)) AS MONTH_NAME

FROM workspace . default . viewshop

DATE (Record2) AS

DATE (Record2) AS

MONTHNAME (Record2) AS

DAYNAME (Record2) AS

TO_DATE (Record2) AS

SELECT COUNT(DISTINCT user_id) AS number_of_users,

user_id,

Record2,

TO_DATE (Record2) AS water_date,

DAYNAME (TO_DATE (Record2)) AS DAY_NAME

CASE

When day_name IN ('Sat', 'Sun') THEN 'a. weekend'

SQL fundamentals

- Exercise
 - Excel exercise
 - Put into database
 - Start coding
- Download ChatGPT
- Create this task in excel / csv
- Upload on database
- Practical exercise - connect to CSV
- Update exercise - join & pdf
- upload to GitHub
- deployment

Combining all code

```

SELECT DISTINCT gender
FROM - userprofile
    
```

Select Distinct

(Duplicate / Data check)

```

SELECT COUNT (*)
    
```

FROM userprofile

GROUP BY user_id

HAVING COUNT (*) > 1;

write
main
Screen

Check categories

```

SELECT DISTINCT gender
    
```

FROM userprofile

Cleaning the gender column

```

SELECT DISTINCT
    
```

CASE WHEN

WHEN

WHEN

else gender

END AS sex

FROM userprofile

o/c

How to Build DATA PORTFOLIO

11/12/21

Portfolio - is showcasing your skills

- Project plan on canvas
- mind
- Deckboards
- Github
- Data Science Projects
- Integrated Learning

Link to your website - need to be sent to Recruiters in a digital Present you need to use a Brand, Start Position yourself. DATA BRAND

ISINGHTEARIN (DATA SCIENCE)

- created dashboard - Githubs Integrated
- Version control on Githubs
- mind

Go to Binglearn

Study guide - download

- Go to chat GPT
- Insert the study guide

Prompt: This year I decided to take a data course @ Binglearn where I learned to

data skills, now I want to add to skill that I have learned @ Binglearn as part of my CV

First 4 Topics to Build Portfolio (download)

- DATA AGENTS
- DATA

Additional Resources

The above document contains some of the information that I have learned @ ISinghtearin, I have extracted the information in bullet points and give it to me so that I can add it on my CV as part of my experience

+ download CV of Paste

Write a prompt for me to create a digital CV portfolio

Don't write the CV

Portfolio App - Generate for site

paste it to portfolio

Vo my version

Page 44

Elephant - my - brand - 30 - portfolio - app

use this image to make a professional picture of me for my digital portfolio. Make me to be wearing formal clothes in a suite

Correct Answer

NULL FUNCTIONS

Miro

Put the case study Viewership

Data Rows 1

New Query - Data Exploration

(1) Select *

From - Viewership

2) Select on

From

WHERE

- is an empty space
 - is a blank (no information)
 - missing value
- enable us to deal with empty, missing values

NULL function

Product	Amount
Apple	100
Orange	NULL
Banana	20

Find the Total Revenue
Select Sum(Amt) As Revenue
From Sales

If NULL statement
Select product,
If NULL (Amount) As Amount
From Sales

Product	Amount
Apple	100
Orange	0
Banana	20

Multiple options
use brackets left to right
the order matters

CASESC

(col1, col2, col3, col4 ...)

- is a way to say replace null values by a value from alternative column or specified value
- If the 1st column is NULL check with col2, If null check col3

We commonly use it in our select statement

Customers

cust-id	cust-name	Phone-Number	Address-Number
101	John	07111 1111	NULL
102	Mary	NULL	062 222 2222
103	Peter	NULL	NULL
104	Abdelhadi	071234 5678	082 5432101

Write an SQL query to give me the name of the customers and their contact details

cust-name, contact-number	John	Mary	Peter	Abdelhadi
07111 1111	082 222 2222	0000	071234 5678	

From customers;

Select cust-name
CASESC (phone-number, address-number, 0000)
get me their phone number, if it didn't get me
+ just give me their address if not
replace with 0000

IS NULL - is a way to check if there is a null value in a specific column

- does a null value exist in the column
- we can use it in our WHERE statement

Customers

Cust-id	Cust-name	Phone-number	Address
101	John	071 111 111	NUU
102	Mary	082 222 222	NUU
103	Peter	NUU	071 234 567 / 082 503 2101
104	Abraham		

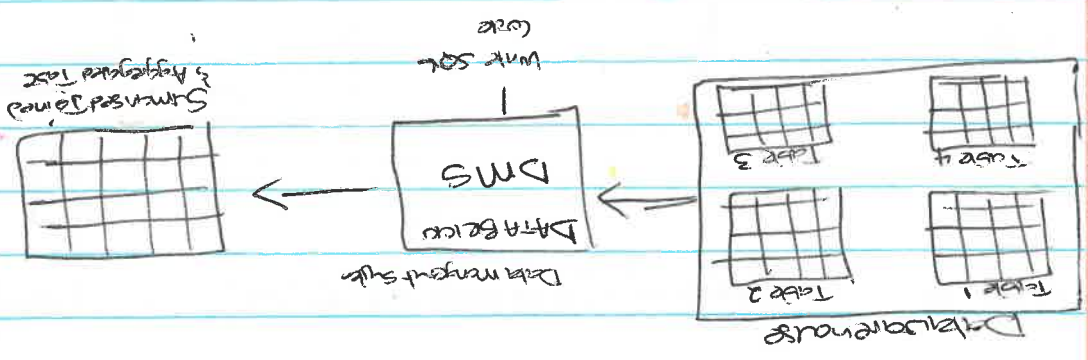
Customer name	Phone number
Mary	NUU
Peter	NUU

SELECT name,
phone-number
FROM customers
WHERE phone-number IS NULL;

Syntax

JOINS

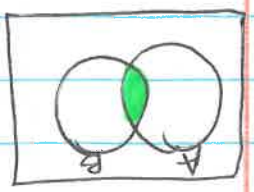
- we want to combine multiple tables into 1 single table
- we join tables based on a common column between the different tables



Tables can be linked to one another based on a foreign key / primary key

How we join the tables

INNER JOIN - joins tables into 1 - returns only rows that have matching values on both tables



Syntax

SELECT col1, col2, cols
FROM TABLE 1 AS A
INNER JOIN TABLE 2 AS B
ON A.common-column = B.common-column

Example INNER JOIN

Customer ID	Customer Name
1	Thabo
2	Sarah
3	Leato
4	John

Customers

Order ID	Customer ID	Product
101	1	Laptop
102	2	Phone
103	2	Headphones
104	5	Tablets

Orders

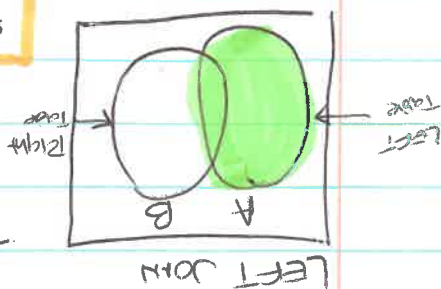
SELECT Customer ID, Customer Name, Order ID, Product
FROM Customers AS A
INNER JOIN Orders AS B
ON A. Customer ID = B. Customer ID

Subquery
Selection
Projection
(filter)
(common column)

Customer ID	Customer Name	Order ID	Product
1	Thabo	101	Laptop
2	Sarah	102	Phone
2	Sarah	103	Headphones

A. Cust ID = B. Cust ID AND A. Order ID = B. Order ID

If Multiple Columns of matching columns



Returns all the rows from the left table and only the matching rows from the right table

I want everything in the left table but from the right table I only take the things that match

SELECT Col 1
Col 2
Col 3
FROM Table 1 AS A
LEFT JOIN Table 2 AS B
ON A. Common Column = B. Common Column

Customer ID	Customer Name	Product
1	Thabo	Laptop
2	Sarah	Phone
2	Sarah	Headphones
3	Leato	Null
4	John	Null

SELECT Cust ID, Cust Name, Product
FROM Customers AS A
LEFT JOIN Orders AS B
ON A. Customer ID = B. Customer ID

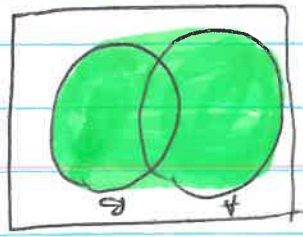
Full Outer Join - It returns all rows matching

all rows that are matching a common column are combined into a single row

It returns all unmatched rows

Where there is no match it will put a null

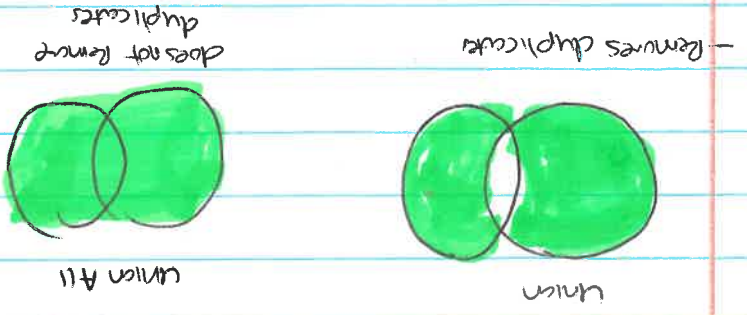
SELECT col1, 2, 3
FROM table 1 AS A
Full Outer Join table 2 AS B
ON A.common column = B.common column



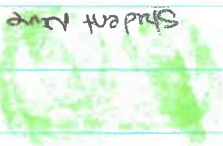
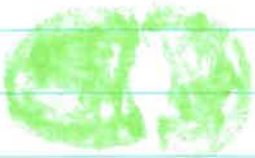
SELECT
Cust ID
Cust name
order ID,
Product
FROM customers AS C
Full Outer Join Orders AS O
ON C.custid = O.custid

UNION AND UNION ALL

- Combines the result of 2 select statements by stacking the rows
- Each The select statements must have the same name of columns
- The columns must be in the same order
- The columns must have the same data types



Custid	1	2	2	3	4	5
Cust name	Travis	Sarah	Sarah	Leah	John	NULL
orderid	101	102	103	NULL	NULL	104
Product	Laptop	Phone	Headset	NULL	NULL	Tablet



SELECT Student-id, Student name
FROM current-student (*)
UNION
SELECT Student-id, Student name
FROM new-students (B);

Student-id	Student Name	Student ID	Student Name
1	Thabo	3	Leah
2	Sarah	4	John
3	Leah	5	Mary

Student ID	Student Name
1	Thabo
2	Sarah
3	Leah
4	John
5	Mary

World of Data

Certification

① Microsoft Certified: Power BI Data Analyst Associate

- Microsoft Azure
- Get started
- Try for free

Go to AI companion (cloud)

Chatgpt

eg what is Microsoft

journey (Azure AI studio)

- Microsoft Azure open AI
- Azure Data Factory
- Data factories (ETL & ELT)
- Azure SQL
- Github

② Microsoft: Fundamentals certification

③ AWS Certification - Data Analytics

- Explore training

- Machine Learning / Microsoft Learn

④ Google Cloud Platform - Google cloud

Explain to me what is SCP, AWS & Azure

⑤ Youtube: Power BI - Tricks

⑥ What field in Data Analytics you want to specialise in: Hugging Face

Understanding concept

- Extensions
- Github
- Data studio
- Excel
- planning (micro)

⑦ Pickup Problems

- Problem solving
- Attention to detail
- (Domain expertise)
- critical thinking
- collaboration
- asking questions
- curiosity
- Data driven questions

- Data manipulation
- Cleaning
- Statistics
- Data visual
- Programming
- Database mgmt
- Data integration
- Tools
- version control
- Business intelligent tools

⑧ Platform: openai.com

Cloud ai

⑨ Medium (to stay updated with best things in tech)

Youtube - Google data Analytics (google career certificate)

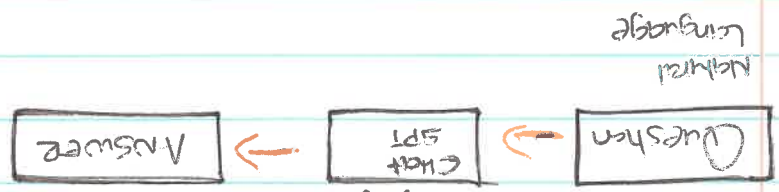
AI for beginners

17/12/23
ChatGPT
Gemma
AI

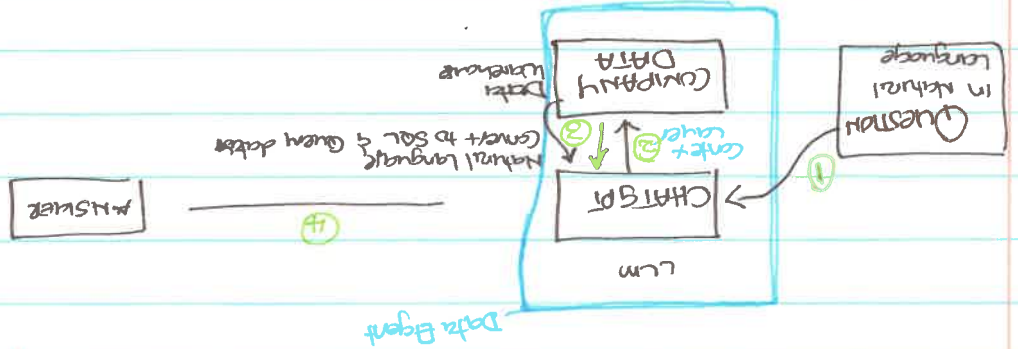
Data Agents (AI)

is a system build within a company, given access to the data so it can generate insights
- After tool Build on large language models
- LLM wait for our prompt to execute instructions

ChatGPT



ChatGPT / LLM - does not know the data within a given company



When we have a data agent, we ask it a question in natural language the question get converted into SQL that SQL script we query that SQL against the database we retrieve the output and give it back to the LLM (ChatGPT) to do analysis for us and send back complete analysis

Data Agents
→ generates you insights on the data

Start asking Question

- What can you do for me
- Show me a restaurant that gives revenue distribution across 2 store location and the top 5 performing products

Data Bricks - Genie Agent

- Click New
- Retail sales
- Click create

Go to Catalogue

- You will see the dataset

- All agent is analysis that

Instruction (DATA Agent)

Confirm

Agent different table

Sources

Tables

About

Self description

Coffee shop business Analyst use provides insights & analysis on Bragt coffee Shop and also gives recommendations on Ads to take market to grow the business

Coffee shop business Analyst put the price

General Instructions - Give the guide to data agent to perform in a specific way

- how it must respond

- its role

Example - Query

how it must

- measure it must use

- Build data Agent "Jeff" - evaluate it - output

DATA AGENT Data to be submitted

Retul - Due Monday -

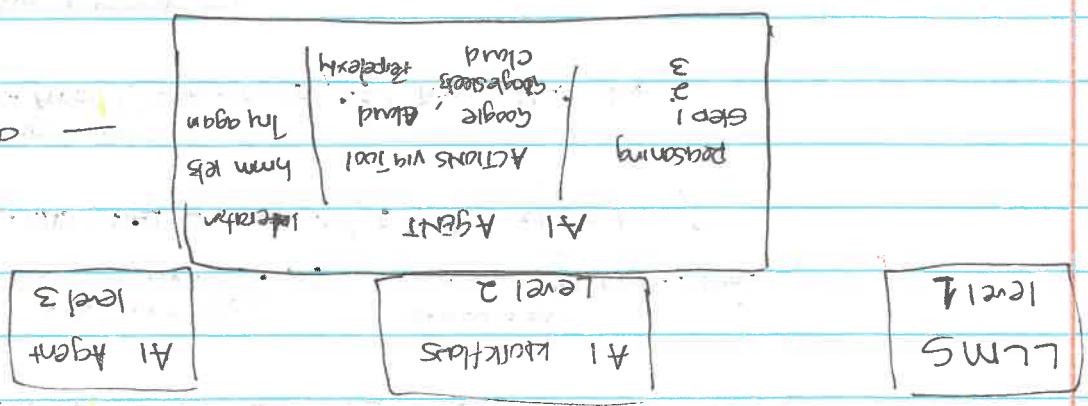
Instructors - Data Agents in Action

Configure
Instructors

copy to action
- I am running a coffee shop & I want to be
Create Agent Instructors based on the data of my
coffee shop for my distributed data agents
Go to Configure
Sample data - copy it & paste on chatGPT

- Data Agent based
- Build data Agent - on your current project
- or the dataset done

187
Financial data
stocks
indexing
(Monika AI)
can read



Artistic
Travel Agency
climb
detectors
mind
to change

(web divpt)

20/12/26 Date Functions

① **Select CURRENT_DATE** AS today-dt; → today's date

② **Select CURRENT_TIMESTAMP()** AS dt-time; → current date & time

③ **Select TO_DATE('2026-07-17')** AS dt; → Ensure that a date is read as a date

④ Convert a date into a string

Select TO_CHAR('2026-07-17', 'yyyy-mm') AS month-id;

⑤ **DATE_DIFF (end-dt, start-dt)**
Basically take end date less start date = value no

Select (TO_DATE('2026-07-05') - TO_DATE('2026-06-01')) AS dt-difference;

Select DATE_DIFF('2026-07-05', '2026-07-01') AS dt-difference

⑥ **Date-ADD (measure, number, days)**
date, new value
you want to add
measure = Day, month, year

Select DATE-ADD (Day, 2026-07-05, 5) AS add-5 days;

Select DATE-ADD ('2026-05-05', -5) AS add-5-days;

⑦ **Select DATE-SUB (date, number, days)**
DATE SUB (date, number, days)
AS remaining-2-days;

Table operations (create as a query)

① Creating a database within a workspace

(CREATE CATALOG IF NOT EXISTS June-Intake; (catalogue)

② **USE CATALOG June-Intake;** (database)

③ **CREATE SCHEMA IF NOT EXISTS students-profile;** (schema)

④ **Drop schema students-profile**
(PARTS OF REPURPOSE TABLE
Catalogue
Schema
June-Intake
Students-profile
Table name profile)

- DATE
- CTE
- Temp Tables
- Joins

Create a Table

CREATE OR REPLACE june - intake . students . profiles (

Student_id INT

name STRING

Surname STRING

age INT

email STRING

Registration - DT DATE) ;

INSERT INTO

june - intake . students . profile

VALUE (101, 'Rocchester, Jones, 28, rocchester@ google . com, '2019-06-05',

102, 'Sue, Magazi, 33

103, 'Casper, Zuma, 73

104, 'Michael, Williams, 41

DELETE function

specific column

We use the delete function to remove a specific row from our data based on a specific condition

DELETE FROM june - intake . students . profiles

WHERE age > 50 ;

UPDATE june - intake . student . profiles

SET age = 25

WHERE student_id IN (101) ;

CTE - You combine it all @ once

Temp Table - You can break it down

- must not include a categorical name
or schema name

Project Descrpt - Grant TV debut

GitHub Repository

Repository

Picture

Repository

New

Repository Name: Grant Exercise

Public

Readme on

2 Files folder

Create a new file

1. Project description / Templates

- Add file - upload file

- Check on Project Descr

Upload image

Commit changes

Commit changes

3 Repository

Copy code

→

Details

Create new

Create folder

Push to url code

Create git folder

Code - Create inside

Create Notebook

Note to Notebook - click on e book

- Right click

- Make to

- Check the folder

- Move it to e

Push

Exercise

Update - hand written
OK - depends → GitHub

After Big code
Excel - DATA VISUALISATION

Graphs
- Pivot table

↓ Vlookup - duplicate Excel
CSV
All rolls

Removing blank
Pivot table
Pivot table Analyse
Change data source
Click on all cell

17 highlight everything
Insert
Pivot table
from table range
New worksheet
OK

apply when columns without - underscore

when separating

closing the table (query)
structure structure query (corresponding in category)

Big Query

With CTE's

① with user_profiles AS (data to clean)
- case splits
- date function
user_profiles table

② Viewership AS (data to clean)
- cleaning data type
- viewership table

③ Select
- column 1
- column 2
- column 3
- column 4

④ From Viewership AS A
⑤ LEFT JOIN user_profiles AS B
⑥ ON A.user_id = B.user_id

With Subquery

Select
From (select code to clean the table)
user_profiles AS A
LEFT JOIN (data to clean)
Viewership table AS B
ON A.user_id = B.user_id

presentation of Dashboard Github

Data studio
Excel desktop
Power BI

Dashboard

- Download the CSV (big code) -
- all rows
- the export is chunked

+ Create - Report

Connect to data

1 Datasets / CSV

2 CSV file upload

3 Create dataset -> Bright TV August
 Schema
 All columns names in files
 listed I want to ok
 Add
 select your file
 continue

Add to Report

Create a Text box

Bright TV viewership Analysis
 Color Black
 Size 49
 Bold
 Background Blue

4 Add a chart

1 Score cards
 shape
 shape
 Drag into metric column

10 odd
 Total Budget hrs
 1823
 438L
 10
 17

Add a chart

Add control (Sliders)

5 Theme & Layout

6 Add Page & Rename (groups)
 Report Page
 Summary Page

Duplicate Page

Page 2

Add chart - graph states

Table
 Dashboard TV Channel
 Metric Score out
 Duration hrs

① Power BI desktop → dashboard

Dashboard planning

When planning a dashboard Ask yourself these questions :

What decision

will you make using the dashboard

How to increase revenue given subscribers base

Who is the dashboard for :

The CEO/Exec team

What heading

What is the 1 thing the dashboard should communicate

- Who are our target mkt is
- Type of content they watch
- When & what drives consumption
- What names are interested in
- What's trending
- Demographics
- What are they applying to
- Where are they applying to

What data

What data supports the headline and what should be left out : Check column names

Supports heading

To be left out

Email, Social media, Event name, Main id

Chart type

line chart

How has a variable changed over time

Bar chart

How do categories compare

What is the part-to-whole breakdown

if 100 pie, how many are white, colored

Stacked bar chart / Pie chart. (< 6 categories) others

Scatter plot

What is the relationship between 2 variables :

What are the single most important numbers
→ KPI card

Dashboard Planning

Understanding Subscribers

Time frame of Report

- Page 1
- Race
 - Age
 - Sub-ID count
 - Region
- Page 2 - writer dashboard
- Day name
 - Day description
 - Time of day

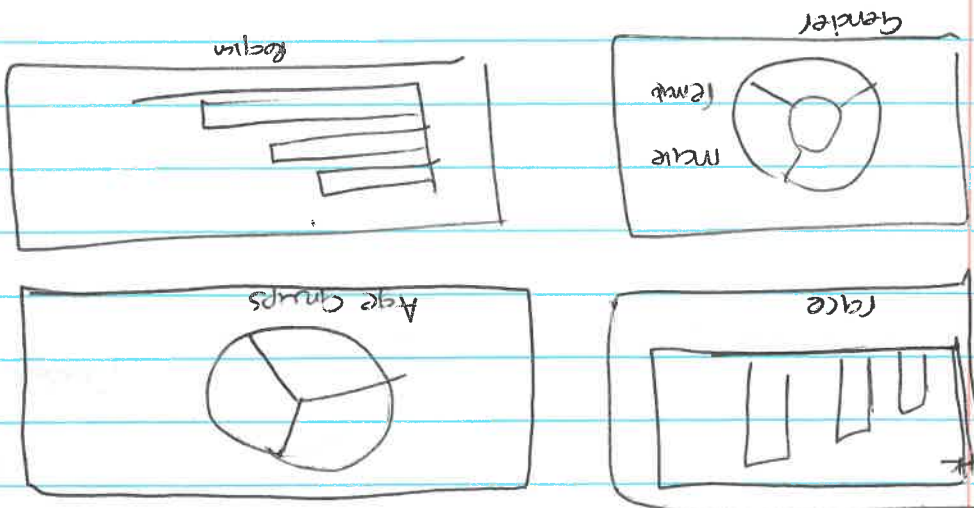
- Page 3 - Screen time by ethnicity
- Duration
 - TV channels

Page

Bright TV Viewership Analysis

Total reach	101k
Total sub	41
Total ms	1523
Aver duration	
TV channels	17

Sub ID count



In census
Which race, gender & region, contributes to most & the least, we understand
Who our subscribers are.

- Excel
- Tableau
- Data studio

Power BI Dashboard
upload data
Click on load

1 Loadable Dashboard
Rajish's link → link
→ link add to receive file
Dashboard
1 Chart SP1

1 Please give me a prompt that I can use for loadable to create a dashboard using this file

2 Paste in loadable

3 If you selected click Rajish

Post on LinkedIn